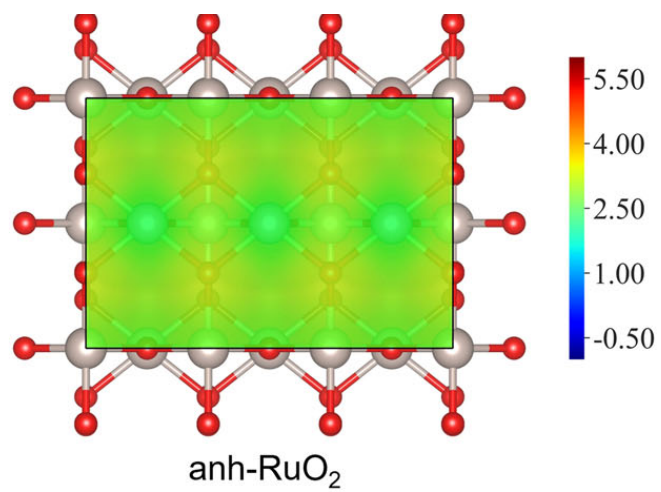
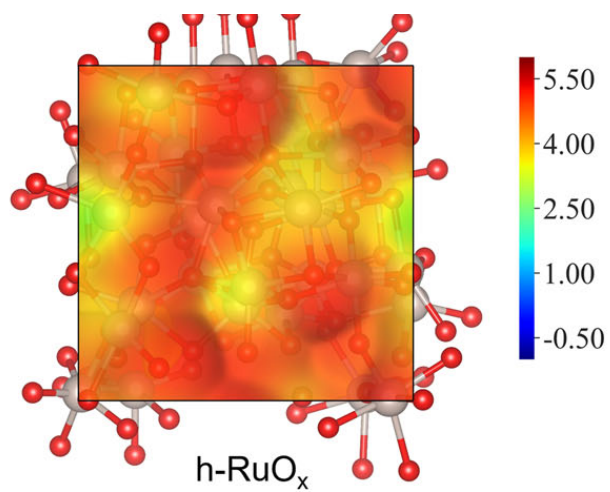


MAIN ARTICLE - REMOVED ORIGINAL CONTENT

Density functional theory (DFT) calculations were conducted to provide a side support of the stability difference observed. Based on the four elementary steps of the acidic OER process³⁴, H₂O molecules



density isosurface from DFT calculation of **h**, the h-RuO_x, and **i**, the anh-RuO₂. The color bar inset illustrates the electrostatic potential ($V_s(r)$ in eV) scale corresponding to different shading colors, and the isosurfaces are plotted at the 0.001 e bohr⁻³ level. The gray ball represents Ru atom, and red ball represents O atom. Source data are provided as a Source Data file.

initially fill oxygen vacancies by forming OH^* species, accompanied by proton removal through a one-electron oxidation step^{35,36}. For the h-RuO_x with its amorphous bulk phase, we utilized ab initio molecular dynamics (AIMD) simulations. For the anh-RuO_2 , we selected the (110) surface in this calculation (Supplementary Data 1). We recognize that selecting an appropriate model for DFT theoretical calculations to thoroughly analyze the stability observed in PEMWE is challenging due to the complex, multi-factor electrode reactions occurring within the device, therefore, the results obtained here serve solely as supporting evidence. The absorption energy of OH^* was calculated to be -1.1 eV, compared to -0.3 eV on the anh-RuO_2 , suggesting that water coordination in the bulk h-RuO_x is notably more active. Considering the subsequent generation of O^* species, where lattice oxygen within the catalyst structure facilitates HOO^* formation to regenerate oxygen vacancies, the calculated lattice oxygen vacancy formation energy in anh-RuO_2 (2.9 eV) is much higher than in h-RuO_x (0.4 eV). Figure 4h, i present the electron-density isosurface of the h-RuO_x and anh-RuO_2 , indicating that h-RuO_x exhibits a more positive electrostatic potential compared to anh-RuO_2 , suggesting a greater affinity for the adsorption of negatively charged OH species. This observation aligns with the calculated adsorption energies, further supports the observed activity and stability differences between the two Ru oxide catalysts in PEMWE.

DFT calculations. All DFT calculations for periodic material systems were conducted using the Vienna Ab initio Simulation Package (VASP) with the projector-augmented wave (PAW) method^{37,38}. The exchange-correlation interactions were described by the generalized gradient approximation (GGA) using the Perdew-Burke-Ernzerhof (PBE) functional³⁹. To account for van der Waals (vdW) interactions, the DFT-D3 approach based on Grimme's scheme was applied^{40,41}. A plane-wave basis set with an energy cutoff of 500 eV was employed, and the interaction between atomic cores and electrons was represented through the PAW method. The Brillouin zone was sampled using a $2 \times 2 \times 1$ k-point mesh centered at the gamma (Γ) point for structural relaxation. A vacuum region of ~ 15 Å was introduced to prevent lateral interactions between adsorbates, and the lower half of the slab models was constrained in their lattice positions during relaxation. Structural optimization was performed by fully relaxing dynamic magnetic moments without restrictions until convergence criteria were satisfied, with total energy differences below 1×10^{-6} eV and residual forces under 0.02 eV/Å.

, The electron-

SUPPLEMENTARY INFORMATION - REMOVED ORIGINAL
CONTENT

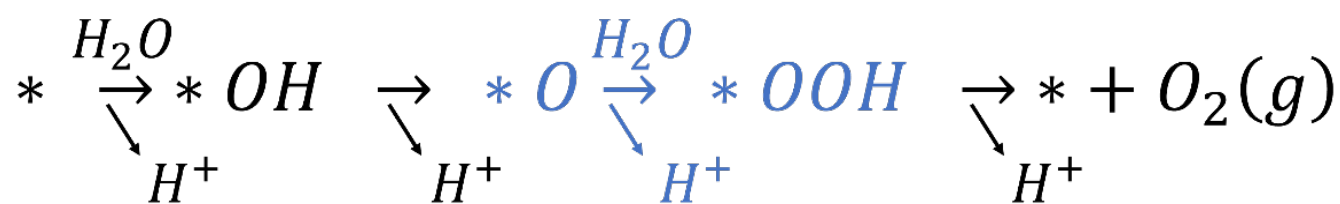
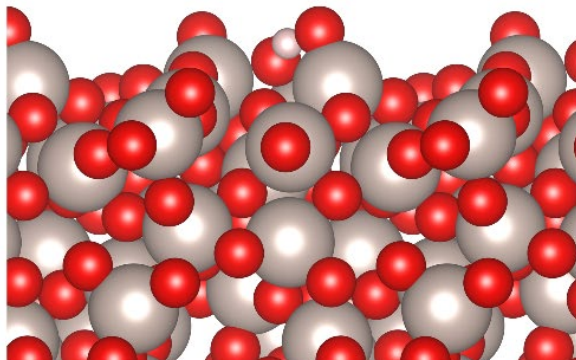


Figure S20. The four elementary steps of acidic OER.

a

h-RuO_x



b

anh-RuO₂

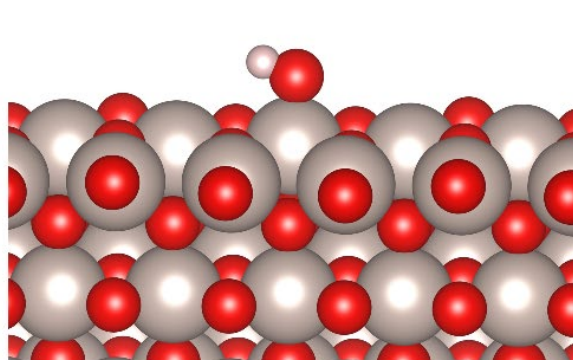




Figure S21. Illustration of the OH* absorption on **a**, the h-RuO_x, and **b**, the anh-RuO₂. The grey ball represents Ru atom, the red ball represents O atom, and the small shiny white ball represents H atom.

Table S4. The calculated lattice oxygen formation energy and OH* intermediate absorption energy based on DFT calculations.

	Lattice O formation (eV)	OH* absorption energy (eV)
h-RuO _x	0.447 eV	-1.106
anh-RuO ₂	2.930 eV	-0.310

RESPONSE LETTER - REMOVED ORIGINAL CONTENT

8. Including more detailed theoretical analysis or computational studies would be helpful in supporting these findings. This would strengthen the argument and provide a more convincing explanation of the observed phenomena.

Peer review p2: assessment of the theoretical calculations

The authors have partially addressed concerns, but the evidence and theoretical calculations for their proposed mechanism remain weak.

2. 2. The simulation part looks rough. Although I understand that the DFT calculation may only qualitatively describe the experimental phenomena, more information should be provided. For example, how is the model of h-RuOx model constructed? Which surface of anh-RuO₂ is considered? Where is the proton that results in the poor stability? How about the free energy profile of the four elementary steps of OER process?

conducted extensive supplementary work on validating the stability and incorporated theoretical analysis to further elucidate the proposed degradation mechanism into the revised manuscript. We believe that these new results provide a clearer demonstration of the

Moreover, in response to the reviewer's suggestion regarding the mechanism of water coordination in relation to stability, we have supplemented additional theoretical calculations to better support the electronic and structural characterizations and to provide deeper insights into this mechanism. These calculations help to elucidate how water coordination impacts the formation energy of oxygen vacancies and how this varies with different Ru electronic structures. This detailed analysis helps to clarify the role that water coordination plays in influencing the stability of Ru oxide catalysts. **We have incorporated these adjustments into both the main text (Figure 4 and analysis) and supplementary**

materials (Figure S20, S21, and Table S4) to ensure that our conclusions are strongly supported by the evidence and offer a clearer narrative.

8. Including more detailed theoretical analysis or computational studies would be helpful in supporting these findings. This would strengthen the argument and provide a more convincing explanation of the observed phenomena.

Response: We sincerely appreciate the reviewer's valuable feedback. **In response, we have included a detailed theoretical analysis based on DFT calculations in the revised manuscript.** We believe these results not only support the observed phenomena but also provide a stronger foundation for the proposed mechanism. We greatly appreciate the reviewer could reconsider the value of this work, given the extensive revisions made in response to all comments.

Density functional theory (DFT) calculations were conducted to provide a fundamental understanding of the stability difference observed. Based on the four elementary steps of the acidic OER process, H₂O molecules initially fill oxygen vacancies by forming OH* species, accompanied by proton removal through a one-electron oxidation step. For the h-RuO_x, with

its amorphous bulk phase, the absorption energy of OH* was calculated to be -1.1 eV, compared to -0.3 eV on the anh-RuO₂, suggesting that water coordination in the bulk h-RuO_x is significantly more active. Considering the subsequent generation of O* species, where lattice oxygen within the catalyst structure facilitates HOO* formation to regenerate oxygen vacancies, the calculated lattice oxygen vacancy formation energy in anh-RuO₂ (2.9 eV) is much higher than in h-RuO_x (0.4 eV). **Fig. 4h** and **4i** present the electron-density isosurface of the h-RuO_x and anh-RuO₂, indicating that h-RuO_x exhibits a more positive electrostatic potential compared to anh-RuO₂, suggesting a greater affinity for the adsorption of negatively charged OH species. This observation aligns with the calculated adsorption energies, further supports the observed activity and stability differences between the two Ru oxide catalysts in PEMWE.

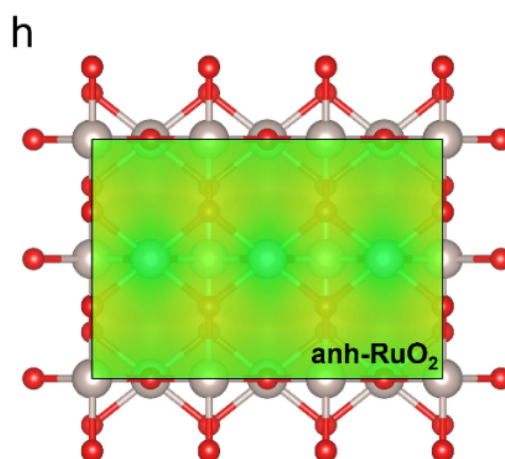
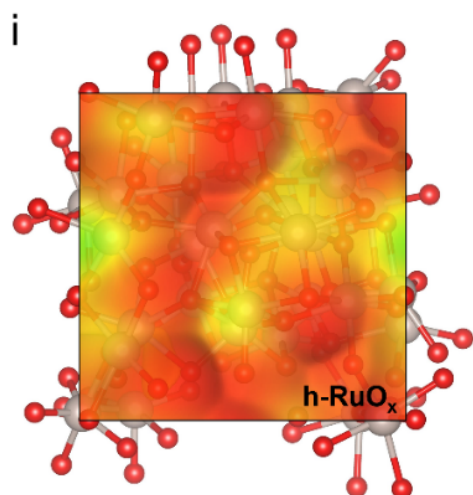
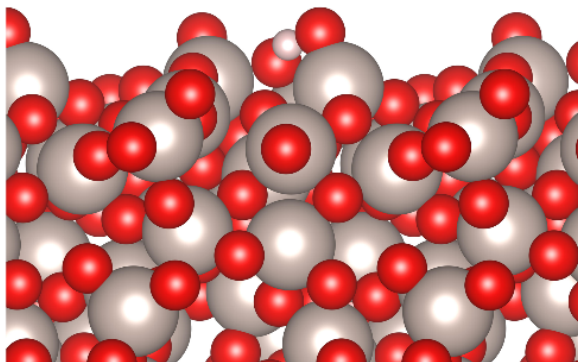


Figure 4. The electron-density isosurface of **i**, the h-RuO_x, and **h**, the anh-RuO₂. The color bar stands for the scale of the electrostatic potential ($V_s(r)$ in eV) and the isosurfaces are plotted at the $0.001 \text{ e bohr}^{-3}$ level.

a

h-RuO_x



b

anh-RuO₂

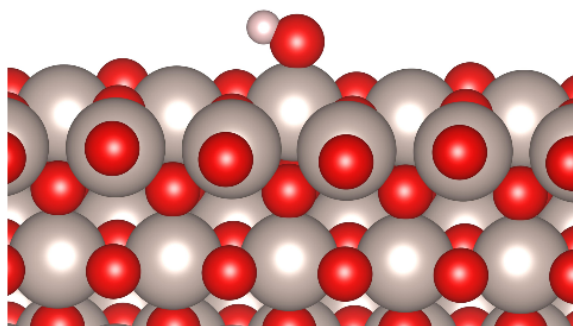


Figure S21. Illustration of the OH* absorption on **a**, the h-RuO_x, and **b**, the anh-RuO₂.

Table S4. The calculated lattice oxygen formation energy and OH* intermediate absorption energy based on DFT calculations.

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2. The simulation part looks rough. Although I understand that the DFT calculation may only qualitatively describe the experimental phenomena, more information should be provided. For example, how is the model of h-RuO_x model constructed? Which surface of anh-RuO₂ is

considered? Where is the proton that results in the poor stability? How about the free energy profile of the four elementary steps of OER process?

Response: Thank you very much for the reviewer's thoughtful feedback and suggestions to strengthen the theoretical calculations, which have significantly enhanced the foundation of our work. We are grateful for the opportunity to provide further details on our approach as suggested by the review, and relevant explanation has also been made to the revised manuscript.

Regarding the reviewer's concern on how the model of h-RuO_x is constructed, we utilized ab initio molecular dynamics (AIMD) simulations. Starting from the structure of crystal RuO₂, we initiated a heating process up to 2500 K. This high-temperature treatment allowed the system to explore a wide configurational space, facilitating the formation of the h-RuO_x structure. Subsequently, we cooled the system back to 300 K to stabilize the newly formed phase. This approach captures the dynamic nature of the h-RuO_x model and provides a realistic representation of its structural characteristics.

In response to the question about which surface orientation of anh-RuO₂ we considered, we selected the (110) surface for our calculations. This orientation was chosen due to its well-documented catalytic activity and alignment with experimental observations, making it a particularly relevant model for investigating the catalytic reactions and surface phenomena of RuO₂.

To address the reviewer's query on the location of the proton that influences stability, our findings indicate that h-RuO_x functions as a mixed proton and electron conductor, inherently incorporating Ru-OH speciation (noted as RuO_x(OH)_y). In OER process, the catalyst can exhibit pseudocapacitive behaviour involving the redox and intercalation of proton, follow the reaction $RuO_x(OH)_y + \delta H^+ + \delta e^- \leftrightarrow RuO_{x-\delta}(OH)_{y+\delta}$ under OER potentials. In this case, intercalated OH dissociates, establishing separate conduction paths for bulk protons and electrons. Conversely, this sort of pseudocapacitive behaviour is much reduced on Ru sites fully occupied by lattice oxygen in anh-RuO₂, which is also facilitated by the theoretical evidence from DFT calculations that anh-RuO₂ exhibited higher formation energies for oxygen vacancies and lower OH affinities at these sites.

We are grateful for the reviewer's understanding that DFT calculations often qualitatively complement experimental data. Thus, we organized this work, mainly focused on evaluating RuO₂ catalyst stability in practical PEMWE systems, including long-duration high-current testing at 1 A cm⁻² and 4 A cm⁻². We acknowledge the reviewer's point that a comprehensive theoretical examination would provide a stronger foundation for our observations. However, given the primary focus of our study, we have addressed the DFT calculations specifically within the revised manuscript: "We recognize that selecting an appropriate model for DFT theoretical calculations to thoroughly analyze the stability observed in PEMWE is challenging due to the complex, multi-factor electrode reactions occurring within the device, therefore, the results obtained here serve solely as supporting evidence."

In this study, we primarily used commercially available RuO₂ catalysts; however, as part of our ongoing efforts to design and synthesize catalysts independently, we have identified some key findings. Specifically, the RuO₂ particles we synthesized after going through an additional hydrothermal treatment can exhibit very unstable behaviour akin to h-RuO_x, while after an additional calcination at 350 °C, the stability can be greatly improved to similar to that of anh-

RuO₂ in PEMWE operation. Since we can closely control synthesis parameters for these custom catalysts, our approach enables a more precise investigation of their structural features, which is ideal for systematic theoretical analyses. These tailored conditions lay a strong groundwork for in-depth DFT studies, and we plan to further investigate the free energy profiles of the four elementary steps in the OER process in our future work, as suggested. We sincerely appreciate the reviewer's insightful guidance and look forward to further collaboration to strengthen and refine these theoretical analyses, ultimately enhancing the depth of our findings.